

## Homework 8. Flight planning activity.

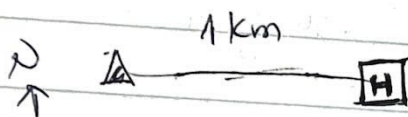
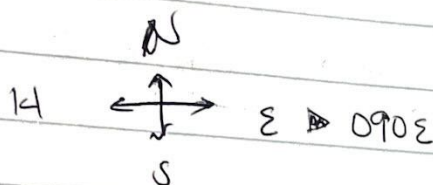
aircraft S-500 quadcopter.

mission : Crop health monitoring flight.

field dimensions : 0.5 km - 500m  
X  
400m

$$\text{CRAFT to field} = s = \frac{D}{T} \quad T = \frac{D}{s} \quad \frac{1000m}{10m/s}$$

$$= 100s \quad 1m40s.$$



- 1
- Terrain  $\Rightarrow$  If it is a plain sight land and so, it's very safe to fly there. altitude is also better. 100 m AGL

- Populated areas

- No the population of the area isn't an overpopulated one.

- Around lake area

• There is quite a plain land with very few buildings.

- There are mostly people around and not animals and they stay at least 700m from the airport.

## 8. Flight Conditions

\* Elevation : 779m

\* Runway : 10/28 : 2325m  
15/33 : 1372m

The elevation of Chileca is approx. 779 meters AGL

The Runway Orientation & Lengths

It is oriented on  $100^\circ$  runway 10 &  $280^\circ$  Runway 28 and its length is 2325m

The other 15/33 Runway is oriented along  $150^\circ$  (Runway 15) and  $330^\circ$  degrees (Runway 33) with a length of 1372m.

## 2. Thoughts on safety to fly.

⇒ Yes it is very safe to fly on this airport's 2 runways why?

Firstly the first runway with length 2,325m can accommodate a wide variety of aircrafts, sizes & weights, which is suitable for commercial aircrafts.

The elevation has a slight impact on aircraft performance during takeoff as it may require a very long distance to achieve lift at higher altitude



The second runway with shorter runway is most suitable for smaller aircraft or drones and light planes.  
With all this the airport seems well-equipped to handle variety of aircraft thus making it efficient and safe.

### 3. Calculations on a scheduled flight at 3PM to morrow.

Details

Current conditions

Today

Temperature

:  $21^{\circ}\text{C}$

Temperature

:  $22^{\circ}\text{C}$

Wind

:

12 mph (10.4)

Wind

:

11 mph (9.58 kts)

Gust

:

23 mph

Gust

:

23 mph 19.9 kts

Is it precipitating

:

No

if precipitating

:

No

Visibility

:

10 miles (16 km)

Visibility

:

11 miles 17.7 km

- Based on the information provided for Chileka Int. airport we can say that it is very safe to fly to and from this airport because all the weather conditions are very okay for flight.

Visibility at 16 km is very excellent for flight and having no precipitation is also an added advantage. The wind is relatively good speed and the temperature is also favourable to fly from.

- The above is also the same for the next day as the temperatures between the 2 days do not differ that much.

Q5. Kind penalty for flying at 3pm  
 $\Rightarrow$  3 minutes since there is 19 kts

Q6. How long will it take to complete the mission?

First let's calculate the horizontal travel to field  
1000 meters from original point @ 10 m/s

$$\begin{aligned} \text{Time} &= \text{Distance} / \text{speed} \\ &= 1000 \text{ m} / 10 \text{ m/s} \\ &= 100 \text{ s} \quad \text{ii)} \end{aligned}$$

$$\begin{aligned} \text{Time to climb to an altitude of 120 m} \\ &= 60 \text{ s} \quad \text{ii)} \end{aligned}$$

$$\begin{aligned} \text{Time to do the survey of the field} \\ \text{Distance} &= \text{Time} / \text{speed} \\ \text{Distance} &= 2 * (0.5 \text{ km} + 0.4 \text{ km}) \\ &= 1.8 \text{ km} (1800 \text{ m}) \end{aligned}$$

Aircraft maneuvers at 5 m/s

$$\begin{aligned} \text{Time taken} &= \text{Distance} / \text{speed} \\ &= 1800 \text{ m} / 5 \text{ m/s} \\ &= 360 \text{ s} \quad \text{iii)} \end{aligned}$$

$$\begin{aligned} \text{Time to descent from 120 m altitude} \\ &= 60 \text{ s} \quad \text{iv)} \end{aligned}$$



v) Time to travel back from field.

$$\begin{aligned} \text{Time} &= \text{Distance} / \text{speed} \\ &= 1000\text{m} / 10\text{m/s} \\ &= 100\text{s} \end{aligned} \quad \dots v)$$

$$\text{Total flight time} = \sum (i, ii, iii, iv, v)$$

$$\begin{aligned} &= 100\text{s} + 60\text{s} + 360\text{s} + 60\text{s} + 100\text{s} \\ &= 680\text{s} \end{aligned}$$

$$= 680\text{s} / 60\text{s/m}$$

$$= \underline{\underline{11.33 \text{ minutes}}}$$

$\therefore$  Yes the mission can be completed in 20 minutes as it has taken about 11 minutes so.

This makes the battery capacity remaining of ~~more~~ about 8.67 minutes thereby making the endurance ~~more~~ reliable. With the wind penalty of 3 minutes, then the final total flight time should be ~~14.3~~ 14.33 minutes.